

A REALIZATION OF F_4 FOR LITT PROBLEM NO. 13

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ABSTRACT. I construct a complex abelian fourfold B and an integral surface $X \subset B$ whose full convolution Tannaka group is the simple group of type F_4 . Starting from the Fano surface of lines on a smooth cubic threefold with an Eckardt point, I quotient its Albanese variety by the elliptic curve parametrizing the lines through that point. The resulting map contracts precisely this elliptic curve and satisfies

$$Rf_*C_F[2] \simeq \delta_X \oplus \delta_0, \quad \chi(\delta_X) = 26.$$

If $H = G(\delta_X)$ and J is the Albert module, then $J|_H \simeq \mathbf{1} \oplus V$ with $\dim V = 26$. Irreducibility on the identity component, the geometry of the Albert cubic, and the subgroup structure of F_4 identify the full group with F_4 . Thus the group of type F_4 is realized in the setting of Litt Problem No. 13.

1. INTRODUCTION

For a complex abelian variety A , convolution of constructible complexes induces a neutral Tannakian structure on the quotient of the category of perverse sheaves by the negligible objects. If $X \subset A$ is an integral subvariety, its perverse intersection complex δ_X therefore determines an algebraic group $G(\delta_X)$. Litt Problem No. 13 asks whether a simple group of type

$$G_2, \quad E_7, \quad E_8, \quad F_4$$

can occur in this way [10].

This paper treats the F_4 case. Recent work characterizes, under natural hypotheses, the smooth subvarieties that have exceptional simple Tannaka groups; the quotient surface constructed here is singular [5].

Theorem 1.1. *There exist a complex abelian variety B of dimension four and an integral surface $X \subset B$ such that*

$$G(\delta_X) \simeq F_4.$$

Here $G(\delta_X)$ is the full convolution Tannaka group, not merely its connected component or its connected derived group.

The construction is geometric. Let $Y \subset \mathbf{P}^4$ be a smooth cubic threefold with an Eckardt point p , let $F = F(Y)$ be its Fano surface of lines, and let $E \subset F$ be the elliptic curve parametrizing the lines through p . The Albanese variety $A = \text{Alb}(F)$ has dimension five, and the Albanese embedding can be normalized so that $E \subset A$ is an elliptic abelian subvariety and

$$G(\delta_F) \simeq E_6$$

with its 27-dimensional Albert representation. I then set

$$B = A/E, \quad X = q(F) \subset B.$$

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The restriction $f = q|_F$ is birational and contracts precisely E . The semismall decomposition theorem gives

$$Rf_*\mathbb{C}_F[2] \simeq \delta_X \oplus \delta_0.$$

Consequently, if $H = G(\delta_X)$ and J denotes the Albert module, then

$$H \hookrightarrow E_6, \quad J|_H \simeq \mathbf{1} \oplus V, \quad \dim V = 26.$$

The translation stabilizer of X is trivial, so $V|_{H^\circ}$ is irreducible. A Hessian calculation for the Albert cubic shows that the H -fixed vector in J has nonzero norm. Thus H is contained in its stabilizer $F_4 \subset E_6$. Finally, no proper positive-dimensional closed subgroup of F_4 in characteristic zero acts irreducibly on the minimal 26-dimensional module. It follows that $H^\circ = F_4$, and hence $H = F_4$.

All varieties and algebraic groups in the paper are over $\mathbb{C}$, and all constructible complexes have coefficients in $\mathbb{C}$. The notation E_6 refers to the simply connected complex group of type E_6 .

2. TANNAKIAN CONVENTIONS

Let A be a complex abelian variety and let $m : A \times A \rightarrow A$ be addition. The convolution product is

$$K * L = Rm_*(K \boxtimes L).$$

Let $\text{Perv}(A)$ be the category of complex perverse sheaves on A . Perverse sheaves of Euler characteristic zero form a Serre subcategory, and the corresponding quotient is the heart of a rigid symmetric monoidal quotient of the constructible derived category. Every tensor subcategory generated by one object is neutral Tannakian [8].

For an integral subvariety $Z \subset A$ of dimension d , let $j : Z_{\text{reg}} \hookrightarrow A$ be the natural locally closed immersion. I write

$$\delta_Z = \text{IC}_Z = j_{!*}\mathbb{C}_{Z_{\text{reg}}}[d]$$

for the perverse intersection complex, extended by zero to A [1]. The skyscraper sheaf δ_0 is the tensor unit. If P is a non-negligible semisimple perverse sheaf, I denote by $G(P)$ the Tannaka group of the rigid tensor subcategory generated by P . Its tautological representation has dimension

$$\dim \omega(P) = \chi(A, P).$$

For the geometric objects used below the resulting Tannaka groups are reductive.

I will use the following functoriality. If $q : A \rightarrow B$ is a surjective homomorphism of abelian varieties, then Rq_* induces a symmetric tensor functor on the quotient categories, and Tannaka duality gives a closed immersion

$$G(Rq_*P) \hookrightarrow G(P). \tag{1}$$

This is [8, Lemma 13.7].

3. AN ECKARDT CUBIC AND ITS NORMALIZED FANO SURFACE

3.1. A concrete cubic threefold. Consider

$$Y = \{x_0^2x_1 + x_1^3 + x_2^3 + x_3^3 + x_4^3 = 0\} \subset \mathbf{P}^4 \tag{2}$$

and the point $p = [1 : 0 : 0 : 0 : 0]$.

Lemma 3.1. *The threefold Y is smooth and p is an Eckardt point.*

Proof. The five partial derivatives of the defining polynomial are

$$2x_0x_1, \quad x_0^2 + 3x_1^2, \quad 3x_2^2, \quad 3x_3^2, \quad 3x_4^2.$$

They have no common zero in projective space: the last three equations give $x_2 = x_3 = x_4 = 0$, while the first two then force $x_0 = x_1 = 0$. Hence Y is smooth.

The tangent hyperplane at p is $x_1 = 0$, and its intersection with Y is

$$x_2^3 + x_3^3 + x_4^3 = 0.$$

This is the cone with vertex p over the smooth Fermat plane cubic. Thus p is an Eckardt point. $\square$

Let $F = F(Y)$ be the Fano surface of lines on Y . It is a smooth projective surface. Its Albanese variety

$$A = \text{Alb}(F)$$

has dimension five, and every Albanese map is an embedding [2]. The lines on Y passing through p are parametrized by a smooth plane cubic; they define an elliptic curve

$$E \subset F.$$

For a point $s \in F$, let $L_s \subset Y$ be the corresponding line.

3.2. A normalization compatible with the elliptic quotient. Choose first a point $e \in E$ as Albanese base point and write

$$a_0 : F \hookrightarrow A, \quad a_0(e) = 0.$$

Then $a_0(E)$ is an elliptic abelian subvariety of A ; I identify it with E .

Let $\mathcal{T} \subset F^3$ be the incidence variety of ordered coplanar triples of lines whose sum is a plane section of Y . The sum map $F^3 \rightarrow A$ is constant on $\mathcal{T}$ [2, 7]; denote its value by $c \in A$.

Roulleau associates to E a reflection involution $\sigma = \sigma_E$ of F and, for $s \notin E$, a point $\gamma_E(s) \in E$ such that

$$Y \cap \langle p, L_s \rangle = L_s + L_{\gamma_E(s)} + L_{\sigma(s)}. \quad (3)$$

This is the complete decomposition of the plane section into its three lines [11, Section 3.1]. The involution σ fixes E pointwise. Its induced automorphism of A acts by $+1$ on T_0E and by -1 on the four-dimensional quotient T_0A/T_0E [11, Theorem 13]. Hence, for the quotient homomorphism

$$q_0 : A \longrightarrow A/E,$$

the automorphism induced by σ is multiplication by -1 .

Lemma 3.2. *The Albanese embedding can be chosen so that simultaneously:*

- (i) $E \subset A$ is an elliptic abelian subvariety;
- (ii) the sum of every coplanar triple in $\mathcal{T}$ is 0;
- (iii) $G(\delta_F) \simeq E_6$, acting through the 27-dimensional Albert representation.

Proof. For a triple of the form (3), the image of its constant Albanese sum in A/E is

$$q_0(a_0(s)) + q_0(a_0(\gamma_E(s))) + q_0(a_0(\sigma(s))) = x + 0 - x = 0.$$

Thus $c \in E$. Multiplication by 3 on the elliptic curve E is surjective, so there exists $u \in E$ with $3u = c$. Replace a_0 by

$$a(s) = a_0(s) - u.$$

This is the Albanese map obtained from another base point of E . Since $u \in E$, the image of E remains the same abelian subvariety and the quotient map is unchanged. The sum of a

coplanar triple becomes $c - 3u = 0$. Moreover, the automorphism induced by σ fixes u , so this change of base point does not alter the induced action $[-1]$ on A/E .

With this normalization, addition maps $\mathcal{T}$ to the origin. The resulting incidence correspondence gives a nonzero morphism

$$\delta_F^{*3} \longrightarrow \delta_0, \quad (4)$$

which is the geometric construction used in [7, Example 2.2(c)]. Put $G_F = G(\delta_F)$ and $J = \omega(\delta_F)$. Kr mer's Gauss-monodromy calculation identifies the derived group of the identity component and its tautological representation as

$$([G_F^\circ, G_F^\circ], J) \simeq (E_6, J_{27}),$$

where J_{27} is a minuscule 27-dimensional representation [6, Theorem 2]. Under the fibre functor, (4) gives a nonzero G_F -invariant cubic form on J . Its restriction to E_6 is a nonzero scalar multiple of the Albert cubic, whose full stabilizer in $\mathrm{GL}(J)$ is E_6 [7, Theorem 2.1(c)]. Since the tautological representation of G_F is faithful,

$$G_F \subseteq E_6,$$

whereas $E_6 = [G_F^\circ, G_F^\circ] \subseteq G_F$. Hence

$$G(\delta_F) \simeq E_6.$$

□

From now on, $F \hookrightarrow A$ denotes the normalized embedding of Lemma 3.2.

4. THE QUOTIENT SURFACE

Set

$$q : A \longrightarrow B = A/E, \quad X = q(F)_{\mathrm{red}} \subset B, \quad f = q|_F : F \longrightarrow X. \quad (5)$$

Then B is an abelian fourfold and X is an integral surface. Let $i : X \hookrightarrow B$ be the closed immersion. Complexes on X will always be regarded as complexes on B by extension by zero along i ; in particular, the notation Rf_* below includes this convention.

4.1. The contracted curve. The tangent bundle theorem of Clemens–Griffiths identifies the projectivized tangent plane to the Albanese-embedded Fano surface with the corresponding line:

$$\mathbf{P}(T_s F) = L_s \subset \mathbf{P}(T_0 A) = \mathbf{P}^4. \quad (6)$$

Roulleau identifies

$$\mathbf{P}(T_0 E) = \{p\} \subset \mathbf{P}(T_0 A)$$

[11, Proposition 4].

Proposition 4.1. *The morphism $f : F \rightarrow X$ is birational. Its only positive-dimensional fibre is*

$$f^{-1}(0) = E.$$

Proof. By (6),

$$\ker(df_s) \neq 0 \iff T_s F \cap T_0 E \neq 0 \iff p \in L_s \iff s \in E.$$

Thus f is immersive on $F \setminus E$. If an irreducible curve $C \subset F$ is contracted, then at a general point of C its tangent line lies in $\ker(df)$, so $C \subset E$. Conversely, E is contained in the kernel of q and is contracted to 0. Since $q^{-1}(0) = E$ inside A , one also has $f^{-1}(0) = E$. In particular, f is generically finite.

It remains to determine its generic degree. Let

$$\pi : \mathbf{P}(T_0A) \dashrightarrow \mathbf{P}(T_0B)$$

be the linear projection with centre $\mathbf{P}(T_0E) = \{p\}$. Since f is immersive on $F \setminus E$, there is a dense open subset $U \subset X_{\text{reg}}$ over which f is finite and unramified. Shrinking U if necessary, I assume that $U \cap B[2] = \emptyset$.

Fix $x \in U$ and $s, t \in f^{-1}(x)$. After translating tangent spaces to the origins of A and B , the differential of q is the quotient map $T_0A \rightarrow T_0B$. The tangent bundle identity (6) therefore gives

$$\pi(L_s) = \mathbf{P}(T_xX) = \pi(L_t).$$

Consequently, L_s and L_t lie in the same plane

$$\Pi = \langle p, L_s \rangle = \langle p, L_t \rangle \subset \mathbf{P}^4.$$

The scheme-theoretic plane-section decomposition (3) shows that the only lines of Y contained in Π are

$$L_s, \quad L_{\gamma_E(s)}, \quad L_{\sigma(s)}.$$

The middle line passes through p and corresponds to a point of E . Since $x \neq 0$ and $f^{-1}(0) = E$, neither s nor t lies in E . Consequently

$$t = s \quad \text{or} \quad t = \sigma(s).$$

The involution σ induces $[-1]_B$ on the quotient, hence

$$f(\sigma(s)) = -f(s).$$

If the second alternative occurred, then $x = f(t) = f(s) = -x$, so $x \in B[2]$, contrary to the choice of U . Hence every fibre over U has one point, and f is birational. $\square$

4.2. The semismall decomposition.

Proposition 4.2. *There is an isomorphism of perverse sheaves on B*

$$Rf_*\mathbb{C}_F[2] \simeq \delta_X \oplus \delta_0. \quad (7)$$

Moreover,

$$\chi(B, \delta_X) = 26. \quad (8)$$

Proof. By Proposition 4.1, f is a proper birational morphism of surfaces, every fibre has dimension at most one, and the unique one-dimensional fibre lies over a point. Hence f is semismall. The only relevant strata are the open two-dimensional stratum and $\{0\}$.

The exceptional fibre is the irreducible curve E . Roulleau computes

$$E^2 = -3 \quad (9)$$

[11, Proposition 10]. Thus the intersection form on the top homology of the exceptional fibre is the nondegenerate 1×1 matrix (-3) . The semismall decomposition theorem therefore yields exactly one skyscraper summand at 0, proving (7); see [3].

The topological Euler characteristic of a Fano surface is

$$\chi(F) = c_2(F) = 27$$

[2, 11]. Taking Euler characteristics in (7) gives

$$27 = \chi(B, \delta_X) + 1,$$

which proves (8). $\square$

5. THE INDUCED SUBGROUP OF E_6

Let

$$H = G(\delta_X), \quad V = \omega(\delta_X).$$

By (8), $\dim V = 26$.

Since $q \circ (F \hookrightarrow A)$ factors through f , equation (7) gives

$$Rq_*\delta_F \simeq \delta_X \oplus \delta_0. \quad (10)$$

The tensor unit already belongs to every rigid tensor subcategory, so

$$\langle \delta_X \oplus \delta_0 \rangle = \langle \delta_X \rangle.$$

Applying (1) and Lemma 3.2 yields a closed immersion

$$H \hookrightarrow E_6. \quad (11)$$

Let J be the 27-dimensional Albert representation of E_6 . Under (11), equation (10) becomes

$$J|_H \simeq \mathbf{1} \oplus V. \quad (12)$$

5.1. Irreducibility on the identity component.

Proposition 5.1. *The translation stabilizer of X in B is trivial:*

$$\mathrm{Stab}_B(X) = \{0\}.$$

Consequently, $V|_{H^\circ}$ is irreducible.

Proof. Factor f through the normalization of X :

$$F \xrightarrow{h} X^\nu \xrightarrow{\nu} X.$$

The morphism h is proper and birational. Since ν is finite, a curve is contracted by h if and only if it is contracted by f . Proposition 4.1 therefore shows that the exceptional locus of h is precisely E . A proper birational morphism between normal surfaces is an isomorphism away from its exceptional locus; hence

$$h : F \setminus E \xrightarrow{\sim} X^\nu \setminus \{z\}, \quad z = h(E).$$

The point z is singular. Indeed, if it were smooth, then locally near z the morphism h would be a birational morphism between smooth surfaces. Its exceptional curves would arise from a sequence of point blow-ups and would be rational, which is impossible because its only exceptional curve is the elliptic curve E . Since h is an isomorphism away from E , it follows that $\mathrm{Sing}(X^\nu) = \{z\}$.

Suppose that translation by $b \in B$ preserves X . It lifts uniquely to an automorphism $\tilde{t}_b$ of the normalization X^ν . This automorphism must fix the unique singular point z . Since $\nu \circ \tilde{t}_b = t_b \circ \nu$ and $\nu(z) = f(E) = 0$, one obtains

$$b = t_b(0) = t_b(\nu(z)) = \nu(\tilde{t}_b(z)) = \nu(z) = 0.$$

Thus $\mathrm{Stab}_B(X) = \{0\}$.

For every $b \in B$, the support of $t_b^*\delta_X$ is $X - b$. Hence

$$t_b^*\delta_X \simeq \delta_X \iff X - b = X.$$

The equality $\mathrm{Stab}_B(X) = \{0\}$ is therefore exactly the nondivisibility condition for X and for the simple perverse sheaf δ_X in the sense of [4, Sections 2.1 and 3.3]. Moreover, $\chi(B, \delta_X) = 26 > 0$, so δ_X is non-negligible. Proposition 3.3(2) of [4] now implies that its tautological representation remains irreducible on the identity component of the Tannaka group. Thus $V|_{H^\circ}$ is irreducible. $\square$

6. THE ALBERT CUBIC AND THE FIXED VECTOR

Let $N : J \rightarrow \mathbb{C}$ denote the E_6 -invariant Albert cubic norm, and let

$$N_3(-, -, -) : J^{\otimes 3} \longrightarrow \mathbb{C}$$

be its complete polarization. Choose a nonzero vector $v_0 \in J$ spanning the trivial summand in (12). Thus H fixes v_0 .

For $v \in J$, define the symmetric bilinear form

$$B_v(u, w) = N_3(v, u, w)$$

and its associated linear map

$$\beta_v : J \longrightarrow J^\vee, \quad u \longmapsto B_v(u, -).$$

6.1. Hessian ranks on the singular cubic. Write the Albert algebra as $J = H_3(\mathbb{O}_{\mathbb{C}})$, where $\mathbb{O}_{\mathbb{C}}$ is the complex octonion algebra. In coordinates

$$M = \begin{pmatrix} a & \bar{\zeta} & \eta \\ \zeta & b & \bar{\xi} \\ \bar{\eta} & \xi & c \end{pmatrix},$$

the cubic norm is

$$N(M) = abc - a n(\xi) - b n(\eta) - c n(\zeta) + 2 \operatorname{Re}((\xi\eta)\zeta), \quad (13)$$

where n is the nondegenerate octonion norm. The rank-orbit description of the Albert algebra is recalled in [12].

Lemma 6.1. *Let $0 \neq v \in J$ satisfy $N(v) = 0$.*

- (i) *If v has Albert rank one, then $\dim \ker \beta_v = 17$.*
- (ii) *If v has Albert rank two, then $\dim \ker \beta_v = 9$.*

Proof. The rank of β_v is unchanged by the structure group and by multiplication of v by a nonzero scalar. Thus the two cases may be computed at the standard representatives

$$e_1 = \operatorname{diag}(1, 0, 0), \quad e_1 + e_2 = \operatorname{diag}(1, 1, 0).$$

At e_1 , the quadratic part of (13) is

$$bc - n(\xi).$$

It has rank $2 + 8 = 10$, so the polarized Hessian B_{e_1} has a 17-dimensional kernel.

At $e_1 + e_2$, write the diagonal perturbation as (α, β, γ) . The quadratic part is

$$\gamma(\alpha + \beta) - n(\xi) - n(\eta).$$

Its rank is $2 + 8 + 8 = 18$, so $B_{e_1+e_2}$ has a 9-dimensional kernel. □

Proposition 6.2. *The fixed vector v_0 has nonzero Albert norm:*

$$N(v_0) \neq 0.$$

Proof. Because H fixes v_0 and preserves N , the map

$$\beta_{v_0} : J \longrightarrow J^\vee$$

is H° -equivariant. By (12) and Proposition 5.1,

$$J|_{H^\circ} \simeq \mathbf{1} \oplus V$$

The two summands are nonisomorphic and occur with multiplicity one. Since H° is reductive, every H° -submodule of J is a direct sum of some of these two constituents. Hence

$$\dim \ker \beta_{v_0} \in \{0, 1, 26, 27\}. \quad (14)$$

If $N(v_0) = 0$, then the nonzero Albert element v_0 has rank one or two. By Lemma 6.1, the kernel dimension would be respectively 17 or 9, contradicting (14). Therefore $N(v_0) \neq 0$. $\square$

7. IDENTIFICATION OF THE FULL GROUP

Choose $\lambda \in \mathbb{C}^\times$ such that $\lambda^3 N(v_0) = 1$. Then $N(\lambda v_0) = 1$. The group H fixes λv_0 , and the stabilizer in E_6 of a norm-one Albert element is a connected simple group of type F_4 , conjugate to the automorphism group of the Albert algebra fixing its identity element [12]. After conjugating inside E_6 , Proposition 6.2 therefore gives

$$H \subseteq F_4. \quad (15)$$

Under this inclusion, V is the 26-dimensional minimal F_4 -module.

Proposition 7.1. *One has $H^\circ = F_4$.*

Proof. By Proposition 5.1, H° acts irreducibly on V . In particular H° is nontrivial and positive-dimensional.

Assume that H° is a proper subgroup of F_4 . Choose a maximal proper positive-dimensional closed subgroup $M \subset F_4$ containing H° . Any M -invariant subspace of V is H° -invariant, so M also acts irreducibly on V .

Liebeck and Seitz classify the positive-dimensional proper closed subgroups of exceptional algebraic groups that act irreducibly on a minimal module [9, Theorem 1]. In the F_4 row for the 26-dimensional minimal module, Table 1.2 contains no proper positive-dimensional irreducible subgroup in characteristic zero; all exceptional entries in that row require positive characteristic. Since the ground field is $\mathbb{C}$, such an M cannot exist. This contradiction proves $H^\circ = F_4$. $\square$

Combining (15) with Proposition 7.1 gives

$$F_4 = H^\circ \subseteq H \subseteq F_4.$$

Thus $H = F_4$, which proves Theorem 1.1.

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